

Introduction to Prime Numbers

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1 Theorem 1

$A \neq B$, A and B both prime

$$\{1, 2, 3, \dots, B\} = \{A \% B, (2A) \% B, (3A) \% B, \dots, ((B-1)A) \% B\}$$

(as sets)

2 Theorem 2

$A \neq B$, A and B both prime

$$\forall C, \{1, 2, 3, \dots, B\} =$$

$$\{(A+C) \% B, (2A+C) \% B, (3A+C) \% B, \dots, ((B-1)A+C) \% B\}$$

(for Ms. Shannon)

3 Theorem 3

If a number n is even, then any factorization of n into all prime factors will have 2 as one of the factors.

Proof by contradiction:

Suppose $n = l_1 \times l_2 \times l_3 \times \dots \times l_k$ where each factor is not necessarily prime

Consider if each factor l_i was an odd number, i.e. not divisible by 2

Then $l_1 \times l_2$ is odd since any product of odd numbers is also odd

Then also $(l_1 \times l_2) \times l_3$ is also odd since $(l_1 \times l_2)$ is odd and l_3 is odd

is we continue this argument we see than $l_1 \times l_2 \times l_3 \times \dots l_k$ is odd and therefore $n \neq l_1 \times l_2 \times l_3 \times \dots l_k$

Now if n is even, we have shown that it must have a factor that is not odd, suppose that factor, call it l_i , was not 2, but we know it is at least even, if we continue, we can also use the same reasoning to show that because l_i is even, it has an even factor. If one of the factors is even, but not 2, we can repeat the process. With each application of this reasoning we find even factors that are smaller than the original even factor. i If we have not found 2, we will know there is a smaller factor (since any even number larger than 2 necessarily has at least a factor of 2), if we have found 2, we have proven there is a factor of 2. We continue this recursive process until we have our factor of 2

Q.E.D.